



Application of Fractal theory for crash rate prediction: Insights from random parameters and latent class tobit models

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ABSTRACT

The repercussions from congestion and accidents on major highways can have significant negative impacts on the economy and environment. It is a primary objective of transport authorities to minimize the likelihood of these phenomena taking place, to improve safety and overall network performance. In this study, we use the Hurst Exponent metric from Fractal Theory, as a congestion indicator for crash-rate modeling. We analyze one month of traffic speed data at several monitor sites along the M4 motorway in Sydney, Australia and assess congestion patterns with the Hurst Exponent of speed (H_{speed}). Random Parameters and Latent Class Tobit models were estimated, to examine the effect of congestion on historical crash rates, while accounting for unobserved heterogeneity. Using a latent class modeling approach, the motorway sections were probabilistically classified into two segments, based on the presence of entry and exit ramps. This will allow transportation agencies to implement appropriate safety/traffic countermeasures when addressing accident hotspots or inadequately managed sections of motorway.

1. Introduction

Reducing congestion and accidents are the primary goals of any transport agency. These two issues are interdependent on one another as significant improvements to one could result in substantial impacts on the other (Chang and Xiang, 2003). Furthermore, relieving congestion could also reduce travel delays, such that air quality and economic productivity can be improved. The connotation between congestion and road safety is always a debated issue, by transport planners and safety experts. Some studies argue that the increased level of traffic congestion reduces traffic speeds and therefore leads to less severe crashes. While that may be true, there will also be an increase in the likelihood of exposure and the number of potential conflicts in the congested conditions. This may lead to more crashes, although of a less severe nature (Quddus et al., 2009).

Existing studies explore the applications of different statistical models in estimating crash frequencies, crash rates, and injury severities at specific locations. For example, count data models such as Poisson and Negative Binomial models were extensively used to estimate crash frequencies (Shankar et al., 1995; Famoye et al., 2004; Bhat et al., 2014); Tobit model was used to study the accident rates (Anastasopoulos et al., 2008); the Mixed Logit (Moore et al., 2011), the

Ordered Logit and the Ordered Probit models (O'Donnell and Connor (1996), Kockelman and Kweon (2002) were used to predict injury severity. These studies considered road geometric variables (ramps, shoulder width, and gradient), pavement characteristics (roughness, rutting, and friction), environmental factors (rainfall, cross wind speed, and snow) and traffic variables (AADT, posted speed limit and the heavy vehicle proportionality). Traffic variables were considered a proxy for traffic congestion in most of the earlier studies. However, modeling the aggregated crashes at a road segment level with such proxies for congestion may obscure the actual relationships (Quddus et al., 2009). In this regard, some researchers have used disaggregated crash records and a measure of traffic congestion during the accident period (Zheng et al., 2010; Ahmed et al., 2012; Yeo et al., 2013). Nevertheless, studies of this category require extensive traffic and crash data, which can be computationally cumbersome.

Within this study, we apply the Hurst Exponent, a metric from Fractal Theory, as a congestion indicator, as reported by Chand et al. (2017) for crash-rate modeling. We analyze one month of traffic speed data at several monitor sites along the M4 motorway in Sydney, Australia and assess the congestion patterns in terms of the Hurst Exponent of speed (H_{speed}). Finally, we use Random Parameters and Latent Class Tobit models to estimate the effect of congestion (in addition to other

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fundamental variables) on long-term crash rates. The detailed procedure is discussed in the following sections.

2. Methodology

2.1. Fractal theory and the Hurst exponent

Fractal Theory, introduced by Benoit Mandelbrot (1967) is useful for studying irregularities in a time series (Mandelbrot, 1967). Objects that can be viewed with a similar appearance at different magnification scales are called fractals. While classical geometry deals with objects of integer dimensions (1-d lines and curves, 2-d squares and circles, etc.), fractal theory deals with objects of non-integer dimension, called Fractal Dimension. This dimension depends on the complexity of the shape, i.e., a shape with a higher fractal dimension is more complicated or rough, than one with a lower dimension and fills more space (Breslin and Belward, 1999). There are various methods to estimate the fractal dimension of a time series, such as the box-counting method, Hurst exponent and Higuchi method. However, a widely used practice for researchers is to calculate the Hurst Exponent using rescaled range (or R/S) analysis (Hurst, 1951; Bo and Rashed, 2004; Chand et al., 2016). Despite having several applications in many scientific fields, applications of fractal analysis of time series data is rarely explored in the transportation domain. Presently, research is limited to observing the fractal nature of traffic flow variables, rather than looking at potential real-world applications (Chand et al., 2017).

The step-wise procedure of the standard R/S analysis is shown below:

For a time-series, $X = X_1, X_2, \dots, X_n$

(1) Calculate average,

$$m = \frac{1}{n} \sum_{i=1}^n X_i \quad (1)$$

(2) Calculate mean adjusted series Y ,

$$Y_t = X_t - m \quad t = 1, 2, \dots, n \quad (2)$$

(3) Calculate cumulative deviate series Z

$$Z_t = \sum_{i=1}^t Y_i \quad t = 1, 2, \dots, n \quad (3)$$

(4) Calculate range series R

$$R_t = \max(Z_t) - \min(Z_t) \quad t = 1, 2, \dots, n \quad (4)$$

(5) Calculate the standard deviation series S

$$S_t = \sqrt{\frac{1}{t} \sum_{i=1}^t (X_i - m)^2} \quad t = 1, 2, \dots, n \quad (5)$$

(6) Calculate rescaled range series (R/S)

$$\left(\frac{R}{S}\right)_t = R_t / S_t \quad t = 1, 2, \dots, n \quad (6)$$

$(R/S)_t$ is averaged over the regions $[X_1, X_t]$, $[X_{t+1}, X_{2t}]$ until $[X_{(m-1)t+1}, X_{mt}]$ where $m = \text{floor}(n/t)$. The value of t is normally chosen so that it is divisible by n .

(7) Finally, Hurst exponent (H) is the slope of the regression line that estimates the relationship between (R/S) versus t in log-log axes.

The following formula relates the Hurst exponent, H and fractal dimension, D .

$$D = 2 - H \quad (7)$$

For a time-series,

- i A value of H in the range 0.5–1 is indicative of a time series having a long-term positive autocorrelation. A high (or low) value will likely be followed by another high (or low) value in the series, i.e., the future trend is more likely to follow an established trend. For

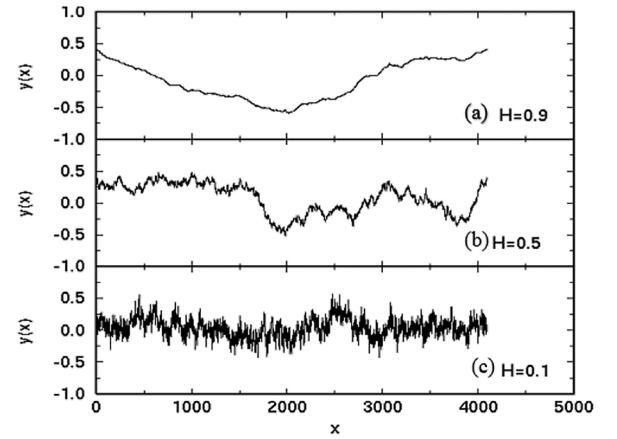


Fig. 1. Demonstration of the Hurst Exponent.

(Source: Cooperative Phenomena Group, n.d.).

example, a very high H value (say $H = 0.9$) means a greater level of determinism (as shown in Fig. 1a) and easily predictable. Large H_{speed} values can be seen after the occurrence of an incident or removal of a bottleneck on a roadway. Incident does not necessarily imply an accident; vehicle breakdown, fire hazard, police pull over, sudden braking and lane changing in dense traffic can result in high H values.

- ii H values close to 0.5 (Fig. 1b) indicates a completely uncorrelated series. The values in the time series are random and potentially indicating Brownian motion. It becomes extremely challenging to predict the future values for such time series.
- iii H value of 0–0.5 suggests the long-term fluctuation between high and low values in the time series. A low H value (say $H = 0.1$) indicates a strong determinism and good predictability despite being volatile (Fig. 1c). A single high value will likely be succeeded by a small value or vice-versa. For example, small H values can be observed for traffic counts on downstream links at signalized intersections, when the measurement interval is shorter than the cycle time of the signal.

Fundamentally, large H values indicate weak dynamics, whereas small H values indicate frequent changes, i.e. high dynamics. From these phenomena, researchers have applied the Fractal Dimension and the Hurst Exponent for incident detection, short-term predictions, and accident warning models. In a recent study, researchers demonstrated that high H_{speed} could be used as a congestion indicator (Chand et al., 2017). They observed high H_{speed} values at several traffic monitor sites during weekdays and evening peak hours. Furthermore, they also observed a strong correlation between high H_{speed} and historical incidents. Reinforcing this premise, it was observed that strong dependence (high H_{speed}) would manifest after the occurrence of an incident. There can be multiple instances in a time-window where a location has high H_{speed} . These occurrences may not be limited to police-reported incidents such as; accidents, breakdowns and police stops, but also unreported incidents such as; abrupt braking and lane changing in dense traffic conditions, which were considered as surrogate safety measures in earlier studies (Gettman and Head, 2003; Lee et al., 2006; Oh and Kim, 2010; Bagdadi, 2013). Therefore, the frequent occurrence of high H_{speed} at a location, can indicate a potentially higher number of crashes.

2.2. Tobit model

There has been an enormous emphasis in past research on factors that determine the frequency of accidents (Anastasopoulos et al., 2008). Using exposure-based accident rates (continuous variable) instead of traditional accident frequencies (count variable) as the dependent variable has significant appeal because accident rates are typically used

in crash reporting, i.e., the number of accidents per 100 million vehicles. The use of accidents per vehicles or accidents per vehicle kilometers has an intuitive appeal in highway safety—providing a standardized measure of relative safety on roadway segments, which is more easily interpreted than number of accidents per time period (Anastasopoulos et al., 2008).

Crash rates that are assessed over a finite period along highway sections can result in numerous segments having either zero or minimal accidents reported during the analysis period. This results in a clump of zeros (20 out of 172 within this study) in the dependent variable, which generates biased and inconsistent parameter estimates of modeled accident rates, by standard ordinary least squares (Anastasopoulos et al., 2008; Washington et al., 2010). Tobit model is applied in these circumstances, where crash rate is considered as a censored dependent variable (Tobin, 1958). Few extensions or variations of the Tobit model are used in the literature. For example, multivariate Tobit (Anastasopoulos et al., 2012b), random parameters Tobit (Anastasopoulos et al., 2012a; Anastasopoulos, 2016; Park and Lee, 2017), Bayesian spatial random parameters Tobit Zeng et al. (2017) and random effects Tobit models Chen et al. (2014) are used to analyze both crash rates and injury severities.

In a Tobit model, there is a latent variable Y_i^* which is observable only above a given threshold (zero in this study). The latent variable Y_i^* can be formulated as:

$$Y_i^* = \beta X_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2) \quad (8)$$

The model assumes that the latent variable, Y_i^* is linearly dependent on the independent vector of variables X_i via the vector of parameters β . Also, there is a normally distributed error term ε_i , with zero mean and constant variance σ^2 across observations, i . Finally, the observable variable Y_i is defined as:

$$Y_i = \begin{cases} Y_i^*, & \text{if } Y_i^* > 0 \\ 0, & \text{if } Y_i^* \leq 0 \end{cases} \quad (9)$$

The likelihood function for Tobit model over N number of observations is as follows:

$$L = \prod_{i=1}^N \left[1 - \Phi\left(\frac{\beta X_i}{\sigma}\right) \right]^{1-I(Y_i)} \left[\Phi\left(\frac{Y_i - \beta X_i}{\sigma}\right) \right]^{I(Y_i)} \quad (10)$$

Where $\Phi(\cdot)$ is the cumulative density function; $\phi(\cdot)$ is the probability density function; σ is the standard deviation of the error term; $I(\cdot)$ is the indicator function which takes value zero if $Y_i = 0$ and 1 otherwise.

The expected value of the dependent observed variable Y_i is:

$$E(Y_i X_i) = \beta X_i \Phi(z) + \sigma \phi(z) \quad (11)$$

where $z = \beta X / \sigma$ and $\Phi(z)$ represent the probability of Y_i being greater than zero, $\phi(z)$ is the standard normal probability density function and σ is the standard deviation of the error term.

2.3. Unobserved heterogeneity

Crashes are complex and less predictable events, making it next to impossible to collate all the data that could determine the likelihood of a highway accident. The absence of relevant explanatory variables can have serious model specification problems for traditional statistical analyses, which can potentially lead to biased and inconsistent parameter estimates, erroneous inferences and inaccurate accident predictions (Mannering et al., 2016). This particular problem is referred to as ‘unobserved heterogeneity’. Researchers will typically adopt statistical approaches such as random parameters models (Milton et al., 2008; Anastasopoulos and Mannering, 2009; Anastasopoulos et al., 2012a; Venkataraman et al., 2013; Anastasopoulos, 2016; Pande et al., 2017), latent class (finite mixture) models (Park and Lord, 2009; Eluru et al., 2012; Zou et al., 2013; Shaheed and Gkritza, 2014; Yasmin et al., 2014), a combination of latent class and random parameters models

(Xiong and Mannering, 2013; Buddhavarapu et al., 2016), Markov-switching models (Malyskina and Mannering, 2009, 2010; Malyskina et al., 2009), and Markov-switching models with random parameters/latent classes (Xiong et al., 2014) to address the problem. The last three approaches are complex processes and computationally cumbersome, as noted by Mannering et al. (2016). In the current study, the Random Parameters and Latent Class Tobit models are used to account for unobserved heterogeneity in crash rate prediction. A brief overview of both the approaches is presented in the following sections.

2.4. Random parameters model

In random parameter models, estimated parameters can vary across observations according to an analyst-specified distribution, typically normal, log-normal, triangular and uniform, to account for the heterogeneity.

The estimable parameter can be viewed as:

$$\beta_i = \beta + \phi_i, \quad i = 1, 2, \dots, N \quad (12)$$

where ϕ_i is a randomly distributed term with mean μ and variance σ^2 . The log-likelihood function (LL) can be written as follows:

$$LL = \sum_{i=1}^N \ln \int g(\phi) P(Y_i^* | \phi) d\phi \quad (13)$$

where $g(\cdot)$ is the probability density function of the randomly distributed term, ϕ . The maximum likelihood estimation procedure of random parameter models is computationally cumbersome (Anastasopoulos et al., 2012a). Therefore, a simulation-based maximum likelihood method is generally employed using Halton draws, which has been shown to provide an efficient distribution of draws for numerical integration (Halton, 1960; Train, 2000; Bhat, 2001).

The RP formulation presented above assumes the same random parameter mean for all the individuals (highway segments). This approach has been widely used in many earlier studies (Anastasopoulos and Mannering, 2009; Venkataraman et al., 2013; Pande et al., 2017). There are few extensions, such as RP models with correlated parameters and RP models with means (and variances) as functions of explanatory variable that are lately applied in accident modeling (Kim et al., 2013; Venkataraman et al., 2014; Mannering et al., 2016; Behnood and Mannering, 2017a, 2017b; Seraneeparakarn et al., 2017). These approaches offer a more general formulation of unobserved heterogeneity. However, these approaches are not within the scope of the present study.

2.5. Latent class model

Having not been applied in existing literature for crash rate modeling, Latent Class Model (LCM) is similar to an RP Model, in the sense that it also accounts for unobserved heterogeneity. LCM has been applied in past studies on crash models that focused on injury severity (Yasmin et al., 2014; Sasidharan et al., 2015). In latent class models, unobserved heterogeneity is accounted by classifying observations into a set of Q different latent classes, with homogeneous variable effects within each group. LC models consist of two components: a class membership model and a class-specific regression model (Vij et al., 2013). In the case of crash analysis, the class membership model formulates the probability that a highway section belongs to a specific latent segment, as some function of the section’s characteristics. Conditional on the class that a section belongs to, the class-specific Tobit regression model evaluates the effect of different explanatory variables on the crash rate. The main advantage of LCM is that it does not impose a distributional assumption on the parameter, thus making it semi-parametric (Greene and Hensher, 2003; Mannering et al., 2016). The estimated parameters are assumed to have identical effects across elements belonging to that segment, but different effects between classes.

The class membership probability for a location i to belong to class q follows a logit model:

$$P_{iq} = \frac{\exp(\theta_q z_i)}{\sum_{q=1}^Q \exp(\theta_q z_i)}, \quad q = 1, 2, \dots, Q \quad (14)$$

where z_i denotes a set of attributes (covariates) that are associated with class membership, and θ_q is a corresponding vector of estimable parameters for the latent segment q . The unconditional probability for a location i is the expectation (over all classes) of the class-specific contributions:

$$P_i = \sum_{q=1}^Q P_{iq} P_{i|q} \quad (15)$$

where $P_{i|q}$ is the expected number of accidents at location i , conditional on latent class q . Finally, the log likelihood is the summation of the logs of all the probabilities:

$$LL = \sum_{i=1}^N \ln P_i \quad (16)$$

The class membership model is rich in terms of interpretation and provides potential applications for policy-makers (Vij et al., 2013) in implementing appropriate crash countermeasures for a targeted segment. Without covariates in a class membership model, one can only infer that there are few distinct classes in the observed data and different effects of variables on each of the classes. However, suggesting crash countermeasures for these classes may be challenging because the class that an observation belongs to will not be known to the analyst.

Both RP and LC models have their drawbacks. RP models require distributional assumptions and are not flexible enough to identify groups of highway segments with shared unobserved heterogeneity. On the other hand, LC models are not flexible to account for unobserved heterogeneity within in the identified latent classes (Mannering et al., 2016). Therefore, few researchers proposed latent class models with random parameters within classes (Xiong and Mannering, 2013; Buddhavarapu et al., 2016). Nevertheless, such models are computationally expensive and also can seriously complicate the estimation routine (Mannering et al., 2016). This hybrid approach is not within the scope of this study.

Furthermore, it should be noted that the results of both the LC and RP models are highly data-specific and therefore, one particular method may not always be necessarily superior to the other (Mannering et al., 2016). This is because unobserved heterogeneity can follow different distribution in different datasets and sometimes LC models may appear to fit data better, and sometimes the RP models may seem to fit better.

3. Study area and data description

Sydney is the largest city in Australia with a population of approximately 4.8 million. Traffic congestion is a major problem in the city, although it has experienced a substantial drop in congestion rankings in recent years. The M4 is a 47-kilometer freeway, predominantly operating with 3-lane carriageways. However, there are few sections configured with 2 and 4-lane carriageways (Fig. 2). There are 16 interchanges along the M4, including a connection to the M7 Motorway. The M4 enables efficient travel between the main regions of western Sydney and services over 100,000 commuters each day. The M4 provides various forms of Intelligent Transportation Systems (ITS), such as; Variable Message Signs (VMS), Variable Speed Limit Signs (VSLs) and Vehicle Detection Systems, also known as loop detectors. Despite the application of ITS on the M4, incidents of different magnitudes still occur along this corridor. These types of incidents include breakdowns, towing, police stops, hazards, and accidents.

The data used in the current study were obtained from 184 monitor sites in the eastbound and westbound directions of the M4 motorway

during the month of February 2013. The westbound is away from the Sydney central business district (CBD), and the eastbound is towards the CBD. The monitor sites are approximately 500 m apart. Each monitor site records traffic data at 30-s intervals, e.g., average speed, traffic flow, and occupancy. Weighted average speed and the total flow across all lanes were calculated. 1-month traffic data were then discretized into 3-h, nonoverlapping windows, starting from 12 AM on 1st of February 2013. Longer time windows, e.g., greater than 4 h, would measure traffic flow to be more deterministic and persistent with H values closer to 1. However, shorter time windows, e.g., 1 h or 30 min, provide an insufficient sample set to estimate an H value (Chand et al., 2017). To reconcile this variability, a 3-h time window is considered in the current study. There are 224 windows (28 days * 8 windows/day) at each monitor site. In other words, we estimate 224 values of H_{speed} at each location. Some monitor sites were not functional throughout the data collection period, and for a few sites, data were either not recorded or exhibited unrealistic values for some part of the day. Such records were not analyzed in the study. This data cleaning exercise resulted in the omission of 12 monitor sites from the analysis. Fig. 3 shows the frequency distribution plots and the summary statistics of H_{speed} in the westbound and eastbound directions. Although H can theoretically take values from 0 to 1, we observed the minimum value as 0.31 in the current dataset.

Long-term crash counts/rates are a commonly accepted measure to classify safe and unsafe road sections (Hauer, 1996). For this study, four-year historical accident data between 2012 and 2015 was obtained from the Roads and Maritime Services (RMS). The data set included information on time, location, duration and accident type. Any accident occurring on the carriageway was attributed to the two nearest monitor sites (one upstream and one downstream). For example, if an accident happened at a 1/4th distance from site A and 3/4th distance from B, then the 3/4th proportion of the accident was attributed to site A and 1/4th to site B. The total number of accidents at each site were then calculated, by summing all these proportions and rounding to the nearest integer. Descriptive statistics are provided in Table 1. Mean values of the observations are provided for the continuous variables, and the percentage of records that fall into each category are provided for categorical and binary variables.

4. Data analysis

4.1. Usability of the Hurst exponent for crash rate modeling

As stipulated in the data section, H_{speed} values were estimated for every 3-h window at each monitor site. Then the percentage frequency distributions were obtained across all 3-h periods at each location. Next, a set of 13 binary variables based on threshold value “ H ” were defined as follows:

High_Hurst_1 = 1; If the Hurst exponent is greater than “ H ”

High_Hurst_0 = 0; Otherwise.

The threshold value H varied from 0.35 to 0.95 in 0.05 increments. This changing value of H produced 13 different binary variables defined as “High_Hurst_0.35” through to “High_Hurst_0.95”. This classification is similar to the “jerk” (rate of change of acceleration) classification approach used in (Pande et al., 2017).

Fig. 4 shows the Pearson’s correlation coefficient between accident rates and percentage of time-windows with High_Hurst_1 = 1, based on the 13 defined thresholds. It may be observed that the correlation coefficient rises, as the threshold value H is increased from 0.35 to 0.95. The increase is noteworthy for the three rightmost threshold values, i.e., 0.85, 0.90 and 0.95. Thus, it can be inferred that sites with a higher proportion of high-value H (a potential indicator of congestion as stipulated earlier) are correlated strongly with long-term crashes. Therefore, the crash rate modeling in this study is based on the threshold, $H_{0.85}$, i.e., the percentage of time-windows with $H > 0.85$.

Next, a Fixed Parameters Tobit model with fundamental variables

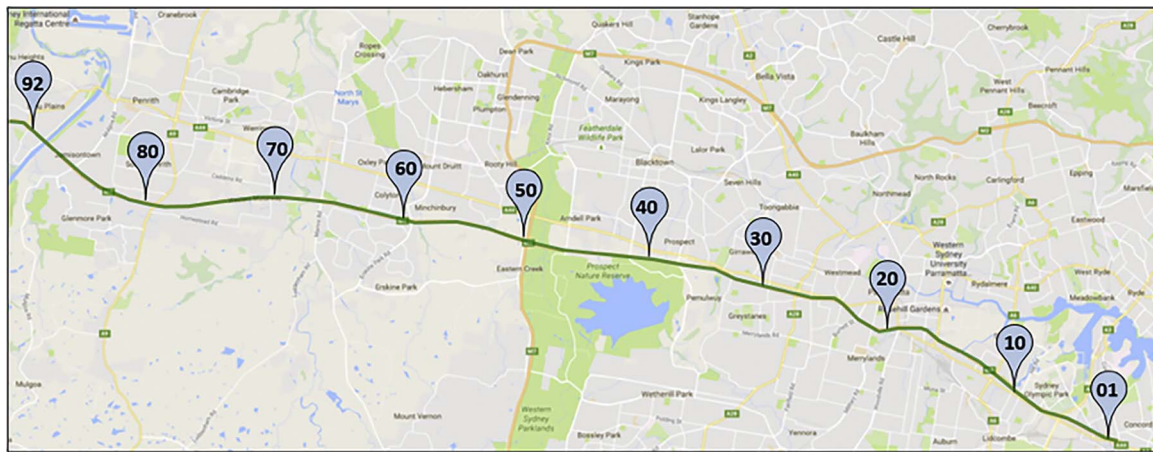


Fig. 2. The M4 motorway in Sydney (with some of the monitor sites labeled).

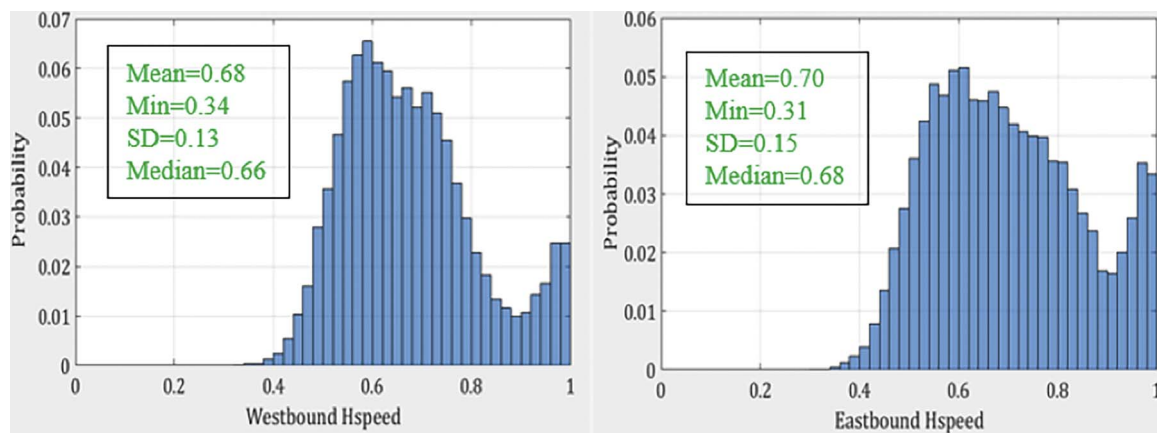


Fig. 3. Frequency distributions of the westbound and eastbound H_{speed} .

(AADT per lane, curve, ramp, bound, the proportion of heavy vehicles) was estimated. $H_{0.85}$ (the percentage of time-windows with $H > 0.85$) is included in the set of explanatory variables, and the model is then re-estimated. In the latter model specification, we observed a significantly better fit in terms of the Akaike Information Criteria (717 compared to 740 in the first model). This signifies the importance of capturing traffic congestion in crash models. However, to account for unobserved heterogeneity and improved model fit, we estimate crash rates using advanced statistical approaches as described in the following section.

4.2. Crash rate modeling

The next step in the analysis is to predict long-term crash rates using random parameters and latent class approaches at the 172 monitor sites under consideration. Simulation-based maximum likelihood with 500 Halton draws was utilized, to estimate the Random Parameters Tobit

(RPT) model. In determining the best model specification for the sample, we estimated several models by changing the mixing distributions (normal, log-normal, uniform and triangular). The variables that are significant at $p = 0.10$ were considered. From all the RPT model specifications explored, the best model in terms of goodness-of-fit outputted the following: $H_{0.85}$ variable with a normally distributed random parameter; entry ramp and exit ramp with lognormal distribution; remaining variables with fixed parameters. The key results of this model are shown in Table 2.

For the latent class model, different model specifications were estimated by: i) changing the number of latent classes and ii) including different explanatory variables (covariates) in the class membership model. Covariates are helpful to assess the probability that an observation belongs to an exclusive segment. Specific variables are only included in the class membership model if they contribute significantly to the segmentation of the motorway sections, i.e., only if the estimated

Table 1
Descriptive statistics of all the considered variables.

Variable	Type	Mean	SD	Min	Max
Crash rate (number of crashes per 100 million vehicles)	Continuous	3.04	3.56	0.00	18.92
Citybound (1 if yes and 0 otherwise)	Binary	48.3% (citybound) 51.7% (outbound)	–	0	1
Horizontal curve (1 for presence and 0 for absence)	Binary	23.2% (curve) 76.8% (no curve)	–	0	1
Ramp (0 = no ramp; 1 = entry ramp; 2 = exit ramp)	Categorical	16.9% (on-ramp) 15.7% (off-ramp) 67.4% (no ramp)	–	0	2
AADT per lane (in 10,000)	Continuous	1.69	0.48	0.65	2.80
% heavy vehicles	Continuous	4.97	1.38	0.86	8.11
$H_{0.85}$ (in percentage of 10) ^a	Continuous	1.50	1.07	0.28	5.41

^a description of this variable in the next section.

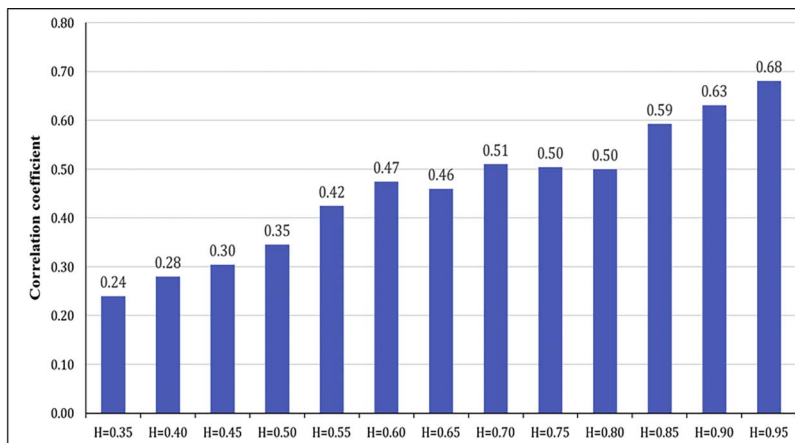


Fig. 4. Correlation coefficient between crash rates and percentage of time-windows with High_Hurst_H = 1 (based on 13 different threshold values of H).

Table 2
Estimation results of RPT and LCT models.

	Random Parameters Tobit (RPT) Model	Latent Class Tobit (LCT) Model	
		Class-1	Class-2
Constant	1.44 (2.31)	14.41 (4.25)	− 0.49 (0.39)
H0.85 (in percentage of 10)	1.89 (4.45)	4.26 (5.90)	0.84 (3.56)
Std. dev. of H0.85	0.61 (3.31)	–	–
AADT per lane (in 10,000)	− 1.57 (− 3.59)	− 6.20 (− 4.23)	0.81 (0.16)
Citybound	− 1.07 (− 3.86)	− 2.86 (− 2.98)	− 0.42 (− 1.26)
Proportion of heavy vehicles	–	− 0.87 (− 1.92)	0.09 (0.63)
Horizontal curve	0.56 (1.96)	–	–
Entry ramp	3.21 (5.28)	Included in the class membership model	–
Std. dev. of entry ramp	2.66 (4.34)	–	–
Exit ramp	4.59 (4.48)	Included in the class membership model	–
Std. dev. of exit ramp	3.40 (3.93)	–	–
Number of observations	172	172	–
LL	− 329	− 313	–
AIC	680	656	–
BIC	714	703	–

	Random Parameters Tobit (RPT) Model	Class Membership component	
		Class-1	Class-2
Constant	NA	− 2.60 (− 3.54)	Fixed
Presence of entry ramp	NA	4.08 (4.09)	Fixed
Presence of exit ramp	NA	4.85 (3.59)	Fixed

t-statistic in parenthesis.

covariate parameters are significant at $p = 0.10$. Similarly, within each class-specific Tobit model, the variables that are significant at $p = 0.10$ for all classes, were considered. Out of all the evaluated LCM specifications, the one with two latent segments and the presence of entry and exit ramps as covariates in the class membership model was found to provide the best fit¹ and physical interpretability², while being parsimonious³. Inclusion of more than four latent classes in the model resulted in convergence issues. Nonconvergence is not an uncommon issue in the latent class models. Previous studies observed that the models with convergence problems are generally those that are badly misspecified (Nylund et al., 2007) and the probability of

nonconvergence increases as the number of latent classes increases. We present the estimation results along with the goodness of fit measures, AIC and BIC for the final selected RPT and LCT models in Table 2. The class membership part for the LCT model is also presented in Table 2.

4.3. Comparison of RPT and LCT models

As stipulated earlier, the results are data-specific as heterogeneity patterns are likely to vary from one dataset to another. Therefore, the inferences made in this study are data-specific as well. In the analyzed dataset, based on AIC and BIC, the LCT model offered a better fit than the RPT model. Upon inspecting the estimation results in Table 2, we observe that the signs of the significant parameters are consistent across the RPT and LCT model frameworks, despite having different magnitudes (due to the underlying parametric distribution assumptions). The results presented in Table 2 for the class membership model in conjunction with the class-specific Tobit model, offer valuable insights into how classes differ, with respect to location types and the varying effect of explanatory variables on each class. In the following section, we compare model estimation results of both frameworks.

¹ The BIC values for 3-class and 4-class models were 707 and 720 respectively, whereas for the 2-class LCT model it was 703. BIC measure resolves the overfitting problem by introducing a penalty term for the number of parameters in the model. The penalty term is larger in BIC than in AIC and the model with the lowest BIC is preferred.

² The 3-class and 4-class models resulted in few insignificant covariates in the class membership model. Covariates are helpful to assess the probability that an observation belongs to an exclusive segment and the insignificance of covariates will not serve the purpose.

³ The number of estimable parameters in 2, 3 and 4-class models were 15, 24 and 33 respectively. As evident, 2-class model is parsimonious.

4.3.1. *H0.85*

The RPT estimation results in Table 2 show that *H0.85* (an indicator of congestion) has a normally distributed random parameter with a mean of 1.89 and standard deviation 0.61. Given this distribution, we can calculate that 99.9% of the locations show an increasing rate of crashes with an increase in congestion. Therefore, according to the RPT model, congestion has a positive effect on the crash rate at almost all the locations in the analyzed dataset. Even in the case of LCT model, *H0.85* has a positive effect on the crash rate in both the classes. However, the effect of congestion in class-1 is more profound than class-2, indicating the congestion severity in this class. The results from the RPT and LCT models concerning the effects of congestion are largely consistent; demonstrating the significance of capturing congestion in crash models.

4.3.2. *AADT per lane*

The relationship between AADT (or AADT per lane) and crash rate in the past studies is slightly ambiguous. Few studies reported a negative relationship between crash rate and AADT (Anastasopoulos et al., 2012a) and few studies reported it as having an inverse U-shaped relation (Hall and Pendleton, 1989; Garber and Subramanyan, 2001; Qi et al., 2007). The inverse U-shape suggests that at low values of AADT, the crash rate increases and then stabilizes/peaks, and finally decreases as AADT increases further. These findings should not be confused with the relation between the “number” of crashes and AADT per lane, which is typically positive and sometimes nonlinear (Abdel-Aty and Radwan, 2000). In the current study, AADT per lane has shown a negative relationship with the crash rate in class-1 of the LCT model. However, in class-2, this variable is insignificant, albeit with a positive sign. It is plausible that the locations in class-1 have a high AADT per lane and are stacked towards the right side of the inverse U-curve and therefore, crash rates decrease with increasing AADT per lane. Locations associated with class-2 probably have lower AADT per lane than that of class-1 and clustered near the peak (slightly towards left) of the inverse U-curve, and so, are insignificant. On the other hand, it is likely that the RPT model considers most of the locations to have AADT per lane on the right side of the curve. It is emphasized once again that all these findings are strictly data-specific.

4.3.3. *Bound*

The city-bound locations were observed to show a negative relation with crash rates in both the RPT and class-1 of LCT frameworks. Similar to AADT per lane, the “bound” variable is found to be insignificant in class-2. The city-bound locations capture drivers predominantly driving towards their workplaces. To contextualize, concentration levels are expected to be higher at these times. Conversely, outbound locations will predominantly capture drivers traveling away from their workplaces and present fatiguing or lowered concentration levels. Thus, resulting in higher crash rates in the outbound direction.

4.3.4. *The proportion of heavy vehicles*

The proportion of heavy vehicles in the traffic stream is found to be insignificant in the RPT model. On the other hand, the variable has a negative effect in class-1 and positive (insignificant) effect in class-2 of the LCT model. It is likely that the negative and positive effects are balanced out, resulting in the insignificance of the variable in the RPT model. It is discussed in the Section 4.3.2 that the class-2 locations are likely to have lower AADT per lane than class-1. Therefore, it is also likely that there are more heavy vehicles in class-1, which could heighten the concentration of light vehicle drivers (Anastasopoulos et al., 2008, 2012a). However, another study reported weak evidence of a negative relationship between crash rate and the proportion of heavy vehicles (Aljanahi et al., 1999). The significance of the variable in class-1 and insignificance in class-2 indicates that the effect of the proportion of heavy vehicles on crash rate can be highly data and site-specific, and heterogeneity considerations are to be made while modeling.

4.3.5. *The presence of a horizontal curve*

The presence of horizontal curves may have heterogeneous effects across highway locations due to unobserved time-varying traffic and weather conditions as well as heterogeneous reactions of drivers to these curves (Mannering et al., 2016). Occasionally drivers will attempt to navigate curves at high speeds, resulting in the tires skidding. This increases the likelihood of a crash eventuating. However, there is an alternate hypothesis that steep curves make motorists to drive cautiously. The RPT model predicts a positive relation between crash rate and the presence of a horizontal curve. Conversely, the LCT model identifies the variable as insignificant in the current data set. This discrepancy could be attributed to the different distribution assumptions and heterogeneity considerations of both the approaches. Further, the characteristics of a curve, such as radius, length, etc. should be captured to precisely estimate the effect of horizontal curves on crash rate.

4.3.6. *The presence of entry and exit ramps*

The RPT estimation results show that the presence of entry and exit ramps have normally distributed parameters. Given the distributions presented in Table 2, we can infer that most of the locations (88.7%) show a positive relationship with crash rate when there is an entry ramp. Similarly, the presence of exit ramps also shows a positive correlation with crash rates at around 91% of the locations. The comparison between the RPT and the LCT models concerning the ramp variable is not simplistic because these variables are included in the class membership component of the LCT model. While the ramp variables are observed to have a direct effect on the crash rate in the RPT model, their effect is indirect in the LCT specification. There will be increased turbulence at ramp locations, due to entering/exiting vehicles and frequent lane changes. These phenomena create traffic bottlenecks/congestion and subsequently, higher crash rates than locations without ramps. From the class membership component presented in Table 2, we observe that the locations with either entry or exit ramps are more likely to belong to class-1 than class-2 (because of their positive signs).

4.4. *A note on the latent segments*

As stipulated earlier, the class membership model formulates the probability that a highway section belongs to a particular latent class. In other words, for each location in the dataset, we obtain probabilities to categorize which class a location belongs to, i.e., class-1 and class-2. Finally, the class membership probabilities for each location are summated to generate the final proportions of each class. Around 1/3rd of the 172 locations were estimated to be in class-1 and the remaining 2/3rd in class-2. Following this, each location was assigned to a single class, based on the highest probability (Magidson and Vermunt, 2004; Nagin, 2005; de Oña et al., 2013). The average observed crash rates for class-1 and class-2 were calculated as 6.33 and 1.49 crashes per 100 million vehicles, respectively. Class-1 exhibited a higher average crash rate than class-2. Apart from congestion, there are significant contributions to crash rate from other explanatory variables in class-1. Conversely, the congestion indicator variable was significant in class-2, despite having a lower magnitude than class-1. This analysis signifies the importance of including congestion metrics in crash prediction models. Furthermore, the findings indicate that the segments belonging to class-1 should be a higher priority (potential hotspots) for transport planners than those in class-2. Greater consideration should be given to ramp locations, due to their increased likelihood of contributing to a high priority segment. Congestion could be reduced at these bottlenecks by implementing traffic management countermeasures such as ramp metering algorithms, variable speed limits and variable message signs, to adequately inform the driver before accessing the ramp.

5. Conclusion

This paper demonstrates the application of the Hurst exponent from

Fractal theory for the prediction of crash rates. Random Parameters and Latent Class Tobit models were estimated to account for unobserved heterogeneity. The best RP model fitted for the dataset had the Hurst variable, the presence of entry and exit ramps with normally distributed parameters. On the other hand, the final latent class model (the best-fit LCT among several explored specifications) suggested the presence of two classes, with the entry and exit ramps as covariates in the class membership model. Segmenting motorway locations into different classes is helpful for transport authorities to implement crash counter-measures. From the explored literature, this is one of the first applications of latent class analysis for “crash rate” modeling. Earlier studies used Multivariate, Random Parameters and Bayesian Spatial Random Parameters Tobit models to predict crash rates. While random parameter models account for unobserved heterogeneity and offer a better fit than a fixed parameters Tobit model, they are time-consuming as the analyst is required to make parametric assumption relating to the distribution of heterogeneity across observations. When considering latent class models, choosing the number of classes and covariates in a class membership component is also cumbersome, but computationally minimal for each specification.

To summarize, this study presents two significant insights into: i) the application of Fractal theory, and ii) the estimation of Random Parameters and Latent Class Tobit models for crash rate modeling. Congestion indication, in terms of the Hurst exponent, is found to be significant in both the high and low priority segments, suggesting the importance of capturing congestion information in crash prediction models. The results of this paper should be treated as exploratory and data-specific because of the different heterogeneity considerations of the RP and LC approaches. Subsequent investigation at a larger scale, and other explanatory variables, such as curvature, pavement quality, weather, traits of motorists, etc. can be explored to reinforce the findings. Despite this, the proposed methodology can easily be applied to newly constructed roads and in developing countries, with insufficient crash data and no standardized police reports. Congestion patterns in terms of the Hurst exponent can be assessed for a limited duration (say one month), for the potential identification of crash hotspots.

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